intercept the localizer, and make an approach to a simulated breakout at an altitude of 61 m (200 ft). At that point, the approach was terminated, and the pilot prepared for the next leg of the route by retuning the radios.

Data

Ground tracks were recorded by an x-y plotter on an aeronautical chart of the New York metropolitan area. Two eight-channel recorders were used for producing time histories of pertinent parameters desired. Localizer error, glide-slope error, and airspeed were recorded on computer printouts for statistical processing. For each run, data points were then selected at predetermined distances from touchdown. These points, in the form of data cards, were processed with a standard statistical computer program to obtain such measurements as standard deviations and mean values. These data were then plotted. Where values are given in both SI and U.S. Customary Units, the measurements and calculations were made in U.S. Customary Units.

RESULTS AND DISCUSSION

En Route Navigation

A composite of the 12 test runs of the complete route structure, 2 for each pilot, is presented in figure 7. The figure also shows the outline of the overall route comprising en route and approach corridors which were 4 n. mi. wide. Even though such a corridor could not be defined for the approach to the Wall Street Heliport because of the proximity of obstructions extending into the airspace, this segment was retained in the simulation to provide a basis for comparison of test results with the VFR experience of NYA. In any event, the pilot had no explicit indication of his position relative to any of the corridor boundaries. Their primary purpose was to provide a qualitative basis for assessing en route navigation performance and for defining the routes in a realistic manner, whereas the minimum en route altitude would provide an obstruction clearance of 152 m (500 ft).

Inspection of figure 7 indicates that no violation of the 4 n. mi. corridor occurred during the en route phase. Although, as shown in the figure, one departure from JFK was inadvertently initiated in the wrong direction, the recovery was still within the corridor. Also, even the procedure turn required on the approach to Newark from the west stayed within the corridor.

Final Approach

Intercept.- Final approach guidance was provided by the MLS. Flight procedures with the MLS were similar to those for a conventional instrument landing system (ILS). The only significant difference was in the course-deviation information; an ILS always